

NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte(Deemed to be University)

II Sem B.Tech (CBCS) Mid Semester Examinations - II, April 2023

EE1001-1 – BASIC ELECTRICAL ENGINEERING

Duration: 1 Hour

Max. Marks: 20

*Note: Answer any **One** full question from **each Unit**.*

Unit – I		Marks	BT*	CO*	PO*
1.	a) State and explain Faradays laws of Electromagnetic Induction.	04	L*2	03	1,2
	b) In a 50 KVA, 1100/220 V single phase transformer the iron and full load copper losses at full load are 350W & 425W respectively. Find (i) Efficiency at full load unity p.f (ii) The load at which maximum efficiency occurs iii) Max efficiency at unity power factor	04	L3	03	1,2
	c) List the various losses in a transformer.	02	L1	03	1,2
2.	a) What is an autotransformer? List its advantages and applications.	04	L1	03	1,2
	b) Derive the emf equation of a single-phase transformer	04	L2	03	1,2
	c) Define self and mutually induced emf.	02	L1	03	1,2
Unit – II					
3.	a) With a suitable diagram explain the construction of synchronous machine.	05	L2	04	1,2
	b) A six pole, lap-wound 230V series motor has the following data: Number of armature conductors = 800, flux/pole = 0.04Wb, total motor resistance = 0.5Ω, iron and friction losses = 1.5kW. If current taken by the motor is 60A, find:				
	i. Total torque				
	ii. Useful torque at the shaft				
	iii. Power output.	05	L3	04	1,2
4.	a) Draw and explain the torque Vs armature current characteristics of DC shunt motor.	03	L2	04	1,2
	b) Derive the emf equation of the DC generator	04	L2	04	1,2
	c) A 12 pole 500 rpm star connected alternator has 48 slots, with 15 conductors/slot, the flux/pole is 0.02 Wb. Assume unity distribution factor and winding factor as 0.97. Calculate line EMF.	03	L3	04	1,2

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
